INCH-POUND
MIL-DTL-18186E(MC)
13 February 2017
SUPERSEDING
MIL-C-18186D(MC)
12 September 1984

DETAIL SPECIFICATION

CROWNS, SERVICE AND DRESS CAP

This specification is approved for use by the United States Marine Corps and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers removable service and dress cap crowns.

1.2 Classification. The service and dress cap crowns are of the following types and sizes as specified (see 6.2):

1.2.1 Types.

a. Type I – Microdenier Polyester, Twill, Laminated, White 2414
b. Type II – (Wool Serge; Green) – DELETED
c. Type III – Polyester/Wool Gabardine, Laminated, Green 2212

1.2.2 Sizes. Service and dress cap crowns are of the following sizes:

Schedule of sizes in inches

6	6½	7	7½
6⅛	6⅝	7⅛	7⅝
6¼	6¾	7¼	7¾
6⅜	6⅞	7⅜	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2000 Lester Street, Quantico, VA 22134 ATTN: SIAT-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL STANDARDS

FED-STD-4	-	Glossary of Fabric Imperfections

COMMERCIAL ITEM DESCRIPTIONS

A-A-50196	-	Thread, Silk
A-A-50199	-	Thread, Polyester Core, Cotton or Polyester-Covered
A-A-52106	-	Cloth, Twill or Plain Weave, Polyester and Polyester Blend
A-A-59963	-	Thread, Polyester

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-10176	-	Cloth, Gabardine: Wool, Polyester and Wool
MIL-DTL-32075	-	Label: For Clothing, Equipage, and Tentage, (General Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1394	-	Provisions for Evaluating Quality of Cap Crowns
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC EP9	-	Visual Assessment of Color Difference of Textiles
AATCC TM8	-	Colorfastness to Crocking: Crockmeter Method
AATCC TM16.3	-	Colorfastness to Light: Xenon-Arc
AATCC TM20A	-	Fiber Analysis: Quantitative
AATCC TM61	-	Colorfastness to Laundering, Home & Commercial: Accelerated
AATCC TM128	-	Wrinkle Recovery of Fabrics: Appearance Method
AATCC TM130	-	Soil Release: Oily Stain Release Method
AATCC TM135	-	Dimensional Changes of Fabrics after Home Laundering
AATCC TM193	-	Aqueous Liquid Repellency: Water/Alcohol Solution Resistance Test

(Copies of these documents are available online at http://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.asq.org/.)

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ASTM INTERNATIONAL

ASTM D1424	-	Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D3574	-	Standard Test Methods for Flexible Cellular Materials–Slab, Bonded, and Molded Urethane Foams
ASTM D3775	-	Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D3776	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193	-	Standard Practice for Stitches and Seams

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials.

3.3.1 Basic material. The basic material for each type crown shall conform to the following:

a. Type I: Cloth, twill: microdenier, polyester, 6.5 ounces, white shade 2414.

b. Type III: Cloth, gabardine, polyester/wool, 10.6 ounces, green shade 2212, conforming to Type II, Class 12 of MIL-DTL-10176.

Type I cloth shall be 100-percent microdenier polyester cloth. The weave shall be 2/2 right hand twill. The shade shall be white shade 2414 and shall match the standard sample. The color of the finished cloth shall match the standard sample when viewed using AATCC EP 9, Option A, under filtered tungsten lamps that approximate artificial daylight (D75 illuminant with a color temperature of 7500±200 K, with illumination of 100±20 foot candles) and shall be a good match to the standard samples under horizon lamplight at 2300±200 K. The cloth shall be lightly brushed as to provide a face and back finish equal to that of the standard sample. The face of the cloth shall be marked “face” at both ends of each roll with letters not less than ½ inch in height. The marking shall be clearly legible in any indelible medium used commercially. Fabric edges shall not be rolled, folded, double scalloped or wavy, and the fabric shall not be skewed, bowed, baggy, ridgy, or wavy. The Type I cloth shall conform to the requirements specified in table I.

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TABLE I. Type I basic material physical requirements.

Material characteristic	Reference paragraph	Material requirement
Weight, ounces/square yard	4.3.1	6 – 7
Fiber identification	4.3.1	Microdenier polyester
Weave	4.3.1	2/2 right hand twill
Yarns per inch (minimum)	4.3.1	
Warp	4.3.1	140
Filling	4.3.1	70
Breaking strength, pounds (minimum)	4.3.1	
Warp		160
Filling		45
Elmendorf tear, pounds (minimum)	4.3.1	
Warp		4.0
Filling		3.0
Dimensional stability, percent (maximum)	4.3.1	
Warp		3.0
Filling		3.0
Colorfastness to (minimum):	4.3.1	
Laundering (1 cycle)		4.0
Light, (40 hours or 170 Kilojoule)		4.0
Crocking		4.0
Wrinkle resistance, rating (minimum)	4.3.1	3.5
Liquid repellency cycle I, temp. III (105 °F), tumble dry:	4.3.1	
As received:		6
1 wash		6
10 wash		5
20 wash		4
30 wash		4
Oil repellency cycle I, temp. III (105 °F), tumble dry:	4.3.1	
As received:		6
1 wash		5
10 wash		5
20 wash		4
30 wash		3
Soil release - mineral oil cycle I, temp. III (105 °F), tumble dry:	4.3.1	
1 wash		3.5
10 wash		3.5
20 wash		3.5
30 wash		3.5
Soil release - corn oil cycle I, temp. III (105 °F), tumble dry:	4.3.1	
1 wash		3.5
10 wash		3.5
20 wash		3.5
30 wash		3.5

3.3.1.1 Laminate for type I basic material. The Type I basic material shall be laminated in order to improve the finished appearance of the cover. It shall be laminated by applying an interlock material to the basic material using a web adhesive. The interlock's yarn size and type shall be 1/70/34 denier textured polyester with a weight per linear yard of 5.5 ounces. The web adhesive shall be a hot melt adhesive product in nonwoven form.

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3.3.1.2 Laminate for type III basic material. The Type III basic material shall be laminated in order to improve the finished appearance of the cover. It shall be laminated by flame bonding foam to the material. The polyester polyurethane foam shall conform to the physical properties in table II.

TABLE II. Polyester polyurethane foam requirements.

Physical property	Requirement	Test method
Density, pound/feet³	1.2+10%	ASTM D3574
Tensile strength, pounds per square inch (minimum)	15.0	ASTM D3574
Elongation, percent (minimum)	300	ASTM D3574
Tear resistance, pounds per linear inch (minimum)	3.0	ASTM D3574
Compression force deflection, psi (minimum)		ASTM D3574
25% deflection	0.3	
50% deflection	0.3	
Retention of tensile strength after 3 hours, 221 °F (105 °C, steam autoclave, % (minimum)	70	ASTM D3574

3.3.2 Lining for crown (type III). The material for lining the Type III crowns shall be a 100-percent polyester cloth conforming to Class 3 of A-A-52106. The color of the lining shall be green shade 2209.

3.3.3 Thread. Thread for seaming, stitching, overedging, and for making buttonholes shall be as specified in table III for each type crown.

TABLE III. Thread.

Operation	Thread	
	Polyester	**Silk**
Seaming, stitching	**Type I (inside and outside); Type III (inside)**	**Type III (outside)**
	Left twist, tex size 24, 3-ply (needle and bobbin)	Left twist, tex size 60, 3-ply (needle and bobbin)
Overedge stitching	**Types I and III**	
	Left twist, soft finish ticket No. 70, 2-ply (needle and bobbin)	
Buttonholes	**Type I**	**Type III**
	Left twist, tex size 24, 3-ply (needle and bobbin)	Left twist, tex size 60, 3-ply (needle and looper)

3.3.3.1 Polyester thread. The polyester thread shall be tex size 24, 3-ply conforming to Type I, Class 1 of A-A-59963 and to the requirements specified in table III. As an alternate, polyester covered cotton thread, tex size range 31-35, 2-ply, conforming to Type II of A-A-50199 may be used.

3.3.3.2 Silk thread. The silk thread shall conform to Type – Machine Twist of A-A-50196 and to the requirements specified in table III.

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3.3.3.3 Color and colorfastness. The color of the thread for Type I crowns shall be white shade (bleached) Cable No. 66050 and the color of the thread for Type III crowns shall be olive drab shade T-1, Cable No. 66023. The colored threads, when tested in accordance with the applicable thread specification (see 3.3.3), shall show good fastness to wet (dry) cleaning, and in addition, the silk thread shall show good fastness to light.

3.3.4 Eyelets and washers. The eyelets and washers shall be brass (0.008+0.001 inch thick) and shall conform to the style and dimensions shown on figure 1. The entire eyelet, or the flange only of the eyelet, shall be enameled to closely match the shade of the basic material. The barrel of the eyelets shall be scored with 6 to 8 scores.

3.3.5 Labels. Each crown shall have a cloth combination identification-size label and a label/tag conforming to 3.3.5.1 and 3.3.5.2. Marking for intermediate packages shall conform to 3.3.5.3.

3.3.5.1 Identification-size label. Each crown shall be provided with combined identification-size label conforming to Type VI, Class 4 of MIL-DTL-32075, except the content shall include the name of supplier [1/], and in addition, the Wool Products Act information shall be included in the label for Type III crowns. The label shall measure not less than 1¾ inches by 1¼ inches with a minimum ¼-inch margin at the top for sewing. The label shall show fastness to laundering and dry cleaning. Instructions for laundering shall be included as indicated:

Type I Instruction Information:

MACHINE WASH WARM, PERMANENT PRESS CYCLE.

TUMBLE DRY, LOW SETTING.

REMOVE PROMPTLY.

DO NOT USE CHLORINE BLEACH.

Type III Instruction Information:

SPOT CLEAN ONLY.

DRY CLEANING NOT RECOMMENDED.

3.3.5.2 Label/tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper used for the tags shall be standard bleached sulfate having a basis weight of 100 pounds with a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener, clearly legible and readable by scanner. The bar coding element shall be a 13-digit National Stock Number (NSN). There shall be a 12-digit Universal Product Code (UPC) assigned for all NSNs by the government. The initial UPC shall appear beneath the code. The bar codes for the NSN and UPC shall be medium to high density and shall be located so that they are completely visible on the item when it is folded and packaged as specified. The label's location shall cause no damage to the item.

3.3.5.3 Marking. When specified (see 6.2), intermediate packages shall be marked in accordance with the requirements of MIL-STD-129.

[1/] Size of characters for name of supplier shall be minimum 6 point (approximately 1/16 inch).

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3.4 Design. The crown shall be the Marine Corps design shown in figure 2 having a one-piece crown, two front and two back quarters, and a two-ply one-piece headband. One buttonhole shall be at each side of headband for attaching crown to cap frame. Buttonholes shall finish parallel to and ⅜±1/16 inch from lower edge of headband with the purling on the outside; center of buttonholes shall be 5¾±⅛ inches from center of crown. There shall be two metal eyelets on each side and one on front; the front eyelet shall be centered on the front seam; the side eyelets shall be properly positioned from side seam and top and bottom edges of quarters. The center of each side eyelet shall be 1±⅛ inch from its corresponding side seam, and in horizontal alignment. The center of all eyelets shall be midway between the upper and lower edges of the quarters. All raw edges shall be overedge stitched, and the crown of Type III fully lined. Both Type I and III crowns shall be laminated in order to improve the finished appearance of the cover. The crowns are designed to properly fit the cap frame and crown support with which they are intended to be used (see 6.1).

3.5 Patterns. When specified (see 6.2), standard patterns shall be furnished to the supplier by the Government. Unless otherwise specified in this document, the standard patterns shall provide a ¼-inch allowance for all seams. Standard patterns shall show size, seam allowance, direction lines for cutting, and are marked or notched for proper cutting, placement, and assembly of the component parts.

3.5.1 List of pattern parts. The crowns shall be cut from the materials specified herein in accordance with pattern parts as specified in table IV.

TABLE IV. Pattern parts.

Item	Pattern nomenclature	Piece name	No. of cut parts
	Material: basic material with laminate		
1	Quarters, front	C18186-QUARTR FRNT	2
2	Quarters, back	C18186-QUARTER BCK	2
3	Crown	C18186-CROWN	1
4	Headband	C18186-HEADBAND	1
	Material: Cloth, polyester		
5	Lining, crown (Type III)	C-18186-CRWN LINING	1

3.6 Construction. The crowns shall be manufactured in strict accordance with the operations and the stitch, seam, and stitching types specified in table V.

3.6.1 Manufacturing requirements. The crowns shall be manufactured in accordance with the operations and the stitch, seam, and stitching types specified in table V.

3.6.1.1 Stitches, seams, and stitching. All stitching shall conform to ASTM D6193. The crowns shall be constructed in accordance with the stitch types and stitches per inch specified in table V. Whenever two or more stitch or seam types are specified for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the crown. The number of stitches per inch as specified in table V indicates the minimum and maximum number permitted. Overedge or serge stitching shall be done on serging machines with the knife attachment properly set to trim the raveled edge without cutting material.

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TABLE V. Stitching.

Operation	Stitch type	Seam/stitch type	Stitches per inch
Topstitch, single needle stitch, or straight stitch:	301		
Construct headband: join ends		SSa-1	12-16
Construct headband: fold in half lengthwise and join		OSf-1	12-16
Construct quarters: join at sides Join quarters to crown		SSz-3 (a)	12-16
Construct quarters: topstitch Topstitch quarters and crown		SSz-3 (b)	12-16
Topstitch quarters-headband seam		LSg-2	12-16
Topstitch, single needle stitch, or straight stitch or Single needle chain-stitch:	301 or 401		
Attach lining to crown (Type III only)		SSa-1	6-8
Single needle overedge seam or Serge with double purl on edge	504 or 505	Efd-1	6-8
Safety-stitch: Join headband to quarters	515 or 516 or 519	SSa-2	10-14
Buttonhole	Buttonhole	-	48 per buttonhole including tack

3.6.1.2 Tacking and backstitching. Ends of seams and stitching produced with 301 type stitch when not caught in other seams or stitching shall be backtacked not less than ¼ inch. Thread breaks (all stitch types) shall be secured by stitching back of break not less than ½ inch. When stitch type 401 is used, ends of seams and stitching shall be caught in other seams or stitching, and skipped stitches and thread breaks may be repaired using stitch type 301 back-stitched not less than ½ inch on each side of the skip or break.

3.6.2 Shade marking. The component parts of the crown shall be marked to ensure a uniform shade and size throughout the crown. Any method of shade marking may be used except:

a. Corrosive metal fastening devices.

b. Adhesive type tickets which show discoloration and the adhesive mass which adheres to the material upon removal of the tickets.

c. Any marking medium which would tend to leave a permanent mark.

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3.7 Finished measurements. The finished crown shall conform to the measurements as specified in table VI.

TABLE VI. Finished measurements (in inches).

Size	Circumference [1/]	Height of headband [2/]	Height of quarters at: [3/]			Crown front to back [4/]	Crown side to side [5/]
			Front	Sides	Back		
6	20 7/8	1 11/16	2 3/8	2 1/4	2 1/8	9 3/8	9 1/8
6 1/8	21 1/4	1 11/16	2 3/8	2 1/4	2 1/8	9 5/8	9 3/8
6 1/4	21 5/8	1 11/16	2 3/8	2 1/4	2 1/8	9 3/4	9 1/2
6 3/8	22	1 11/16	2 3/8	2 1/4	2 1/8	10	9 3/4
6 1/2	22 3/8	1 11/16	2 3/8	2 1/4	2 1/8	10 1/8	9 7/8
6 5/8	22 3/4	1 11/16	2 3/8	2 1/4	2 1/8	10 1/4	10 1/8
6 3/4	23 1/8	1 11/16	2 3/8	2 1/4	2 1/8	10 1/2	10 1/4
6 7/8	23 1/2	1 11/16	2 3/8	2 1/4	2 1/8	10 5/8	10 3/8
7	23 7/8	1 11/16	2 3/8	2 1/4	2 1/8	10 7/8	10 5/8
7 1/8	24 1/4	1 11/16	2 3/8	2 1/4	2 1/8	11	10 3/4
7 1/4	24 5/8	1 11/16	2 3/8	2 1/4	2 1/8	11 1/8	11
7 3/8	25	1 11/16	2 3/8	2 1/4	2 1/8	11 3/8	11 1/8
7 1/2	25 3/8	1 11/16	2 3/8	2 1/4	2 1/8	11 1/2	11 3/8
7 5/8	25 3/4	1 11/16	2 3/8	2 1/4	2 1/8	11 3/4	11 1/2
7 3/4	26 1/8	1 11/16	2 3/8	2 1/4	2 1/8	11 7/8	11 5/8
Tolerance:	±1/4	±1/16	±1/8	±1/8	±1/8	±1/8	±1/8

FOOTNOTES:

1/ Measurement for crown circumference shall be taken by laying the crown headband flat and measuring from folded edge to folded edge; this measurement shall be doubled.

2/ Measurement for height of headband shall be taken from the lower finished edge of headband to the lower raised edge of quarters.

3/ Measurement for height of quarters shall be taken at the front, side, and back seams, respectively, from the lower raised edge of quarters to the crown seam.

4/ Measurement for crown front to back shall be taken beginning where the seam for the two front quarter pieces and crown intersect, and end where the seam for the two back quarter pieces and crown seam intersect.

5/ Measurement for crown side to side shall be taken beginning where the seam for the front and back quarter pieces and crown intersect and end on the opposite side where the adjacent front and back quarter pieces and crown intersect.

NOTE: Finished measurements shall be taken as illustrated in figure 3.

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3.8 Fit. Each size crown shall properly fit the corresponding size cap frame and crown support for which it is intended (see 4.5.3.2 and 6.1). Crowns shall fit snugly without distorting either the headband of the frame or the crown wire of the support. Quarters shall have a smooth, rounded appearance without twisting, wrinkles, or other distortions. The crown shall lie smooth and flat without twisting, wrinkles, or other distortions. The headband of the crown shall extend down to the welt on headband of frame without stretching or wrinkling. Buttonholes shall properly engage with posts on frame without twists, wrinkles, or other distortions. The front eyelet of crown shall center on stay of crown support. As an alternate method of measurement, the band circumference may be taken using a 10A gage by fitting the band on the gage and applying pressure past the normal alignment and then relaxing pressure until the alignment is correct at the mark and read measurement.

3.9 Workmanship. The finished crown shall conform to the quality and grade of product established by this specification. It shall be free from any irregular defects that can adversely affect usage or durability, defects that adversely affect form, fit, or function and those defects specified in 4.5.3.1. The occurrence of defects shall not exceed the levels as specified (see 6.2).

3.10 Figures. Figure 1 specifies requirements for the crown's components. Figure 2 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall govern.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When first article is required, inspection shall be made of a completely fabricated crown for conformance to all provisions of this specification (see 3.1). This inspection shall include conformance to the dimensions specified in 3.7 and the examination of defects specified in 4.5.3.1.

4.3 Conformance inspection. Conformance inspection shall include conformance to the dimensions specified in 3.7, the examination of defects specified in 4.5.3.1, and the examination of tests itemized as specified. Unless otherwise specified (see 6.2), sampling for inspection shall be performed conforming to ANSI/ASQ Z1.4.

Each roll in the sample size shall be examined yard-by-yard. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The lot size shall be expressed in yards. The sample unit shall be 5 continuous yards full width finished cloth. The number of rolls from which the sample is to be selected shall be as specified in table VII. For lots which contain fewer than three rolls, each roll in the lot shall be examined. All defects, as defined in FED-STD-4, that are clearly noticeable at normal inspection distance (3 feet), and the defects specified in 4.5.3.1 shall count as a defect.

TABLE VII. Sample size.

Lot size (in yards)	Sample size (in rolls)
1,200 or less	3
1,201 - 3,200	5
3,201 - 10,000	8
10,001 - 35,000	13
35,001 - 150,000	20
150,001 and over	32

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4.3.1 Testing of components. Components and materials shall be tested in accordance with all the requirements of referenced specifications, figures, and standards unless otherwise excluded, amended, or modified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, figures, and standards, testing shall be performed on components listed in table VIII for characteristics noted. All test reports should contain individual values when expressing the final results (see 6.2).

TABLE VIII. Test methods.

Components	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as
Eyelets	Material identification	3.3.4	Standard Commercial	1	Pass or fail
	Color (flange)	3.3.4	Visual	1	Pass or fail
	Scores	3.3.4	Visual	1	Pass or fail
Washer	Material identification	3.3.4	Standard Commercial	1	Pass or fail
Type I basic material	Weight	3.3.1	ASTM D3776		Ounces/square yard
	Fiber identification	3.3.1	AATCC TM20A		
	Weave	3.3.1	Visual 1/		
	Yarns per inch Warp Filling	3.3.1	ASTM D3775		Minimum
	Breaking strength, pounds Warp Filling	3.3.1	ASTM D5034		Pounds, minimum
	Elmendorf tear Warp Filling	3.3.1	ASTM D1424		Pounds, minimum
	Dimensional stability Warp Filling	3.3.1	AATCC TM135, 3 cycles		Percent, maximum

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TABLE VIII. Test methods – Continued.

Components	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as
	Colorfastness to: Laundering (1 cycle), 2/ Light, (40 hours or 170 Kilojoule), 2/ Crocking 3/	3.3.1	 AATCC TM61, option 1A AATCC TM16.3, option A AATCC TM8		Rating, minimum
	Wrinkle Resistance	3.3.1	AATCC TM128		Rating, minimum
	Liquid repellency cycle I, temp. III (105 °F), tumble dry: As received 1 wash 10 wash 20 wash 30 wash	3.3.1	AATCC TM193		Rating, minimum
	Oil repellency cycle I, temp III (105 °F), tumble dry: As received 1 wash 10 wash 20 wash 30 wash	3.3.1	AATCC TM130		Rating, minimum
	Soil release - mineral oil cycle I, temp. III (105 °F), tumble dry: 1 wash 10 wash 20 wash 30 wash	3.3.1	AATCC TM130		Rating, minimum
	Soil release - corn oil cycle I, temp. III (105 °F), tumble dry: 1 wash 10 wash 20 wash 30 wash	3.3.1	AATCC TM130		Rating, minimum

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TABLE VIII. Test methods – Continued.

Components	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as
Polyester polyurethane foam	Density	3.3.1.2	ASTM D3574		Pound/feet³
	Tensile strength	3.3.1.2	ASTM D3574		Pounds per square inch, minimum
	Elongation	3.3.1.2	ASTM D3574		Percent, minimum
	Tear resistance	3.3.1.2	ASTM D3574		Pounds per linear inch, minimum
	Compression force deflection 25% deflection 50% deflection	3.3.1.2	ASTM D3574		Pounds per square inch, minimum
	Retention of tensile strength after 3 hours, 221 °F (105 °C), steam autoclave	3.3.1.2	ASTM D3574		Percent, minimum

NOTES:

1/ One determination shall be made from each sample unit and the result reported as “pass or fail”.

2/ Rated using the AATCC gray scale for color change and AATCC gray scale for staining.

3/ Rated using the AATCC EP9-step chromatic transference scale.

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-1916, except where otherwise indicated in MIL-STD-1394.

4.5 Conformance inspection.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all requirements of references specifications, figures, and standards unless otherwise excluded, amended, or modified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary specifications, figures, and standards, testing shall be performed on components listed in table VIII for characteristics noted. All test reports should contain individual values when expressing the final results (see 6.2).

4.5.2 Sampling for testing. Testing shall conform to the sampling provision tables cited in MIL-STD-1394.

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4.5.3 Examination of the end item. Examination of the end item shall conform to the provisions of MIL-STD-1394.

4.5.3.1 Defects. Defects shall be classified in accordance with table IX.

TABLE IX. Defects.

Examine	Defect	Classification	
		Major	Minor
Material defects and workmanship 1/	Any hole or weakening defect, such as broken, missing, or loose yarn smash, multiple float, drill hole, or loose slub that may develop into a hole.	101	
	Tears, mend, burn, cut, or needle chew.	102	
	Abrasion resulting in a thin or weak area.	103	
	Uneven weaving or incorrect weave.	104	
	Pattern design not equal to the standard sample.	105	
	Incorrect color in any part or the pattern; pattern repeat not equal to the standard sample.	106	
	Multiple flats or skips, ½ inch or more in either warp or filling direction.	107	
Shade	Shade variation within a part or between parts. Thread color not as specified.	108	
	Dye streak; color not as specified.	109	
Cleanness	Any spot or stain clearly noticeable.	110	
	Objectionable odor.		201
	Overall cleanliness including soiling, staining, or embedded matter.		202
	Thread ends not trimmed, or loose thread ends not removed.		203
Components and assembly	Any component or required operation omitted.	111	
	Any component not as specified.	112	
Cutting	Any component part not cut in accordance with specification requirements.	113	

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TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Seams and stitching	Accuracy of seaming:		
	Seams or topstitching twisted, puckered, pleated, not uniform, or otherwise distorted affecting appearance.		204
	End of stitching (when not caught in other seam or stitching) backstitched less than ¼ inch (6.4 mm).		205
	Stitching of thread breaks overlapping less than ½ inch (13 mm) at each end of break.		206
	Any component part caught in an unrelated operation or stitching.	114	
	Gage of stitching:		
	Unevenly gaged, affecting appearance.		207
	Edge or raised stitching sewn too close to edge, resulting in damage to item.	115	
	Open seams. [2/]		
	Less than ¼ inch (6.4 mm).		208
	¼ inch (6.4 mm) but not more than ½ inch (13 mm).		209
	More than ½ inch (13 mm).	116	
	Raw edges. [3/]	117	
	Run-offs; edge or raised stitching (when not resulting in an open seam):		
	¼ inch (6.4 mm) up to ½ inch (13mm) inclusive.		210
	More than ½ inch (13 mm).		211
	Not specified seam or stitch type.	118	
	Loose tension resulting in a loose seam or exposed loose top or bottom thread.		212
	Tight tension (stitches break when normal strain is applied to the lengthwise direction of seam or stitching). [4/]	119	
	Stitches per inch (to be scored only when the condition exists on major portion of seam).		213
Headband	Not finished smooth and flat.	120	
	Twists, pleats, puckers, or other distortion at back seam.	121	
	Not uniform in shape.	122	
	Edge stitching not uniform in width.		214
	Joining seam of headband not in line with back seam of quarters.		215
Quarters	Twists, pleats, puckers, or other distortion at seams.	123	
	Topstitching not uniform.	124	

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TABLE IX. Defects – Continued.

Examine	Defect	Classification	
		Major	Minor
Crown lining (Type III only)	Pleats, puckers, or distortions where the lining joins the crown.		216
	Attachment of lining causing visible stitches on outside of finished crown.	125	
Buttonholes	Not clean cut and well made.	126	
	Not correctly positioned.	127	
	Stitching not securely caught in the fabric.	128	
	Not finishing parallel to and ⅜±1/16 inch from lower edge of headband with the purling on the outside.	129	
	Center of buttonholes not 5¾±⅛ inches from center of front crown.	130	
Eyelets	Not attached from the outside and securely clinched on the inside.	131	
	Attachment causing distortion of, or damage to, the crown or splitting the flange of the eyelet.	132	
	Front eyelet not centered on the front seam.	133	
	Side eyelets not properly positioned. Center of each side eyelet shall be 1±⅛ inch from its corresponding side seam. The center of all eyelets shall be midway between the upper and lower edges of quarters. Side eyelets shall be in horizontal alignment.	134	
Barcode labels/tags	Bar code omitted or not readable by scanner. Human-readable interpretation omitted or illegible.		217
	Bar code not visible on folded, packaged item.		218
	Combined identification and size label not securely caught in stitching, or stitching through the printing.		219
	Combined identification and size label missing, incorrect, or illegible.	135	
	Causes damage to the item.	136	

NOTES:

1/ Material defects are to be classified as indicated only when the condition is one which definitely weakens the fabric or when it is so conspicuously located as to be clearly noticeable. Material defects which are inconspicuous and do not weaken the fabric shall not be classified as defects.

2/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur.

3/ Raw edges not securely caught in the stitching shall be classified as an open seam.

4/ Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.

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4.5.3.2 Fit test. Crowns shall be fit tested against applicable cap frames in accordance with schedule of crown sizes being purchased. The following selected (*) defects are applicable to crowns which improperly fit the corresponding size cap frame. When specified (see 6.2), sample cap frames required for this test may be obtained from the procurement activity or as directed by the contracting officer.

Head Size:

*Defect: Headband of crown fitting frame too loose causing fullness (wrinkles) on band or too tight not permitting crown to be fitted without distorting headband of frame.

Crown Size:

*Defect: Circumference of top crown too small, not permitting crown wire to lie smoothly or distorting crown or quarters affecting appearance. Adjustable crown wires may be adjusted in where necessary.

*Defect: Circumference of top crown too large, not providing snug fit on crown wire or distorting quarters affecting appearance. Adjustable crown wires may be adjusted where necessary.

4.5.4 Dimensional examination. The dimensional examination provisions for the finished crown shall be as listed in 3.7 and conform to MIL-STD-1394.

4.5.5 Examination of packaging. As specified (see 6.2), the examination for compliance with packaging, packing, and marking requirements shall conform to MIL-STD-1394.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers both white microdenier polyester twill dress cap crowns as well as polyester/wool gabardine service cap crowns intended for use with the dress and service cap frame and crown support conforming to MIL-DTL-20268. The referenced cap crowns and frames are intended for wear with Marine Corps Dress and Service Uniforms.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Classification (see 1.2).
c. When first article is required (see 3.1).
d. Requirements for intermediate packages (see 3.3.5.3, 4.5.5, and 6.3).
e. Patterns (see 3.5).
f. Acceptable quality levels (see 3.9).
g. Sampling for inspection (see 4.3).
h. Test report values (see 4.3.1 and 4.5.1).
i. Sample cap frames (see 4.5.3.2).
j. Packaging requirements (see 4.5.5 and 5.1).

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6.3 Intermediate packaging. A white paper label with the following information legibly printed or stamped thereon in black should be inserted within each polyethylene bag in order to permit ready identification:

Nomenclature

Size_____ Stock number_____

Contract number

Contractor

6.4 Subject term (key word) listing.

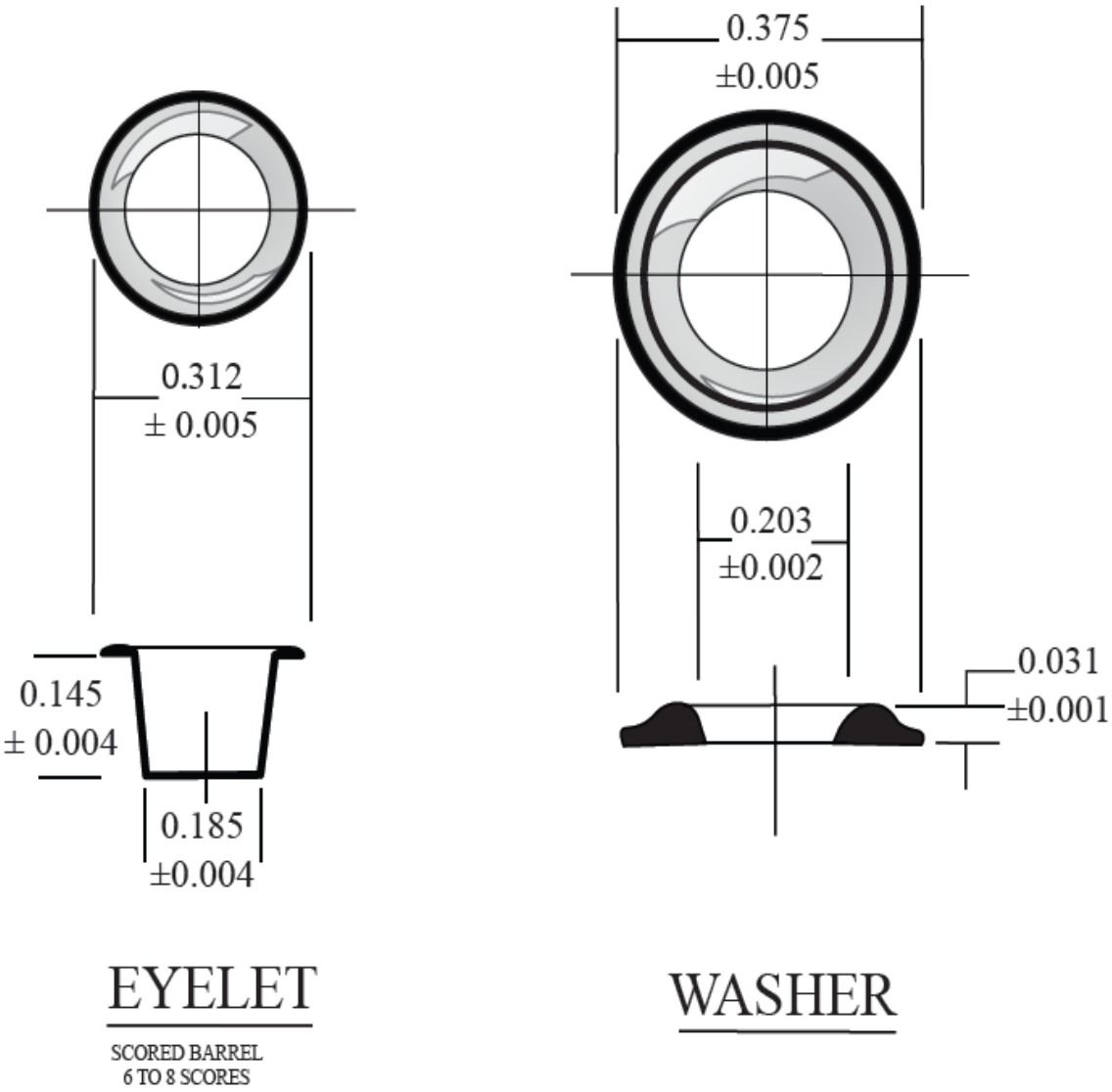
Enlisted

Frame

Officer

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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(All measurements are in inches)

FIGURE 1. Eyelet and washer.

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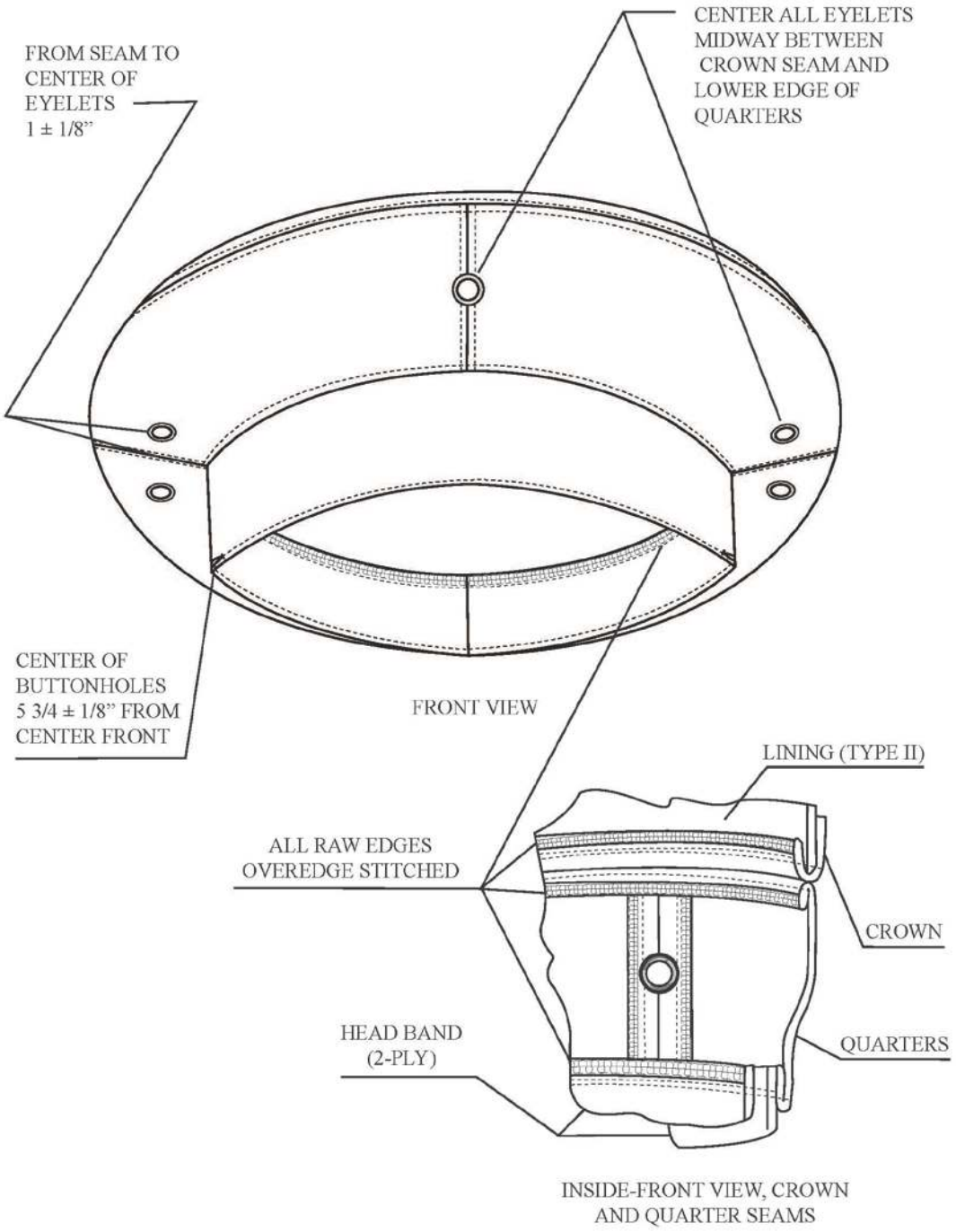


FIGURE 2. Front and inside lining (type III only) views.

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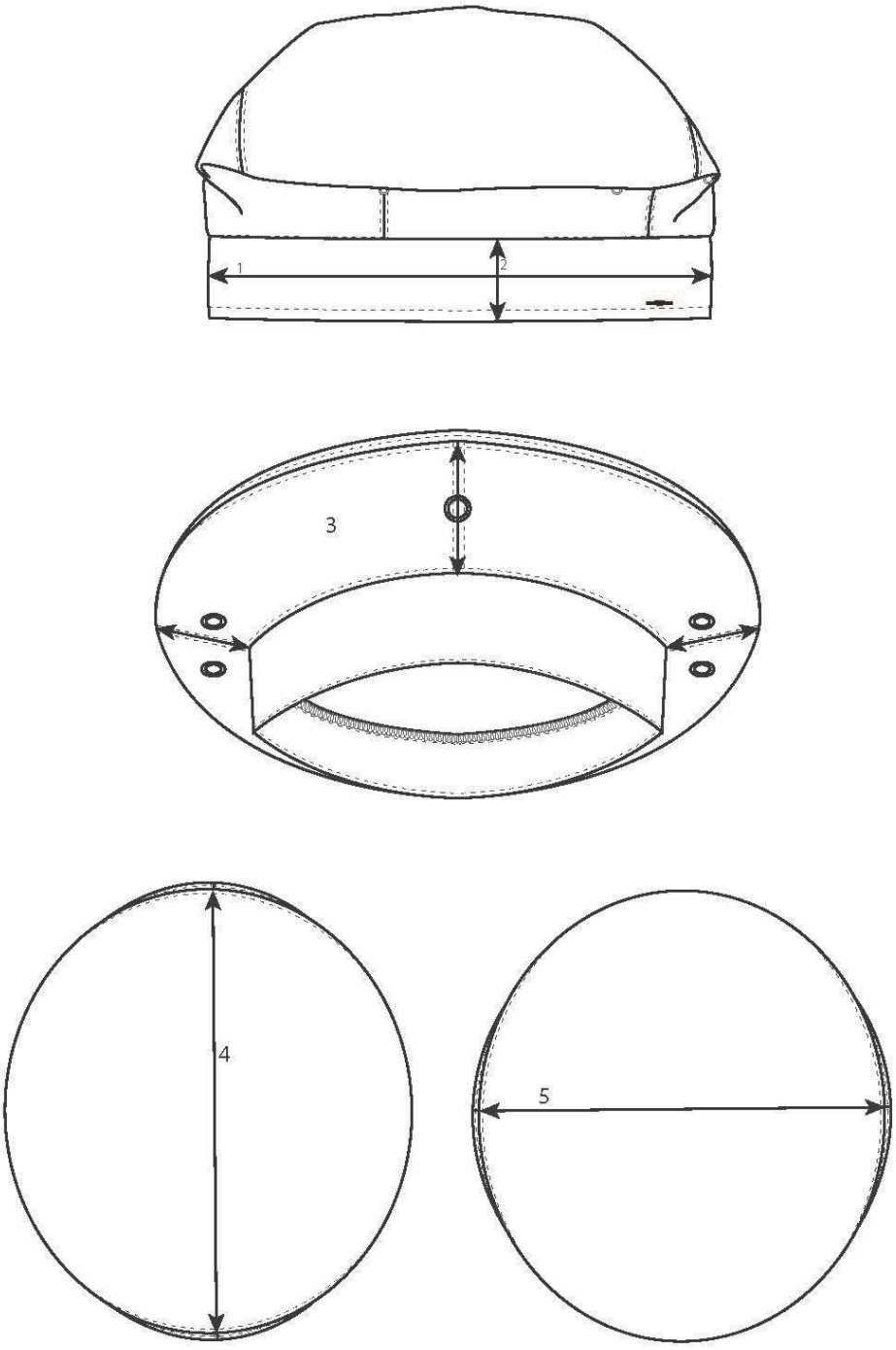


FIGURE 3. Finished measurement instructions.

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